

Iterative Re-estimation of Local Edge-Criticality Metrics across Four Benchmark Networks

NetworkEntropy Research Project

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Abstract

Identifying the edges whose removal fragments a complex network fastest is central to attack-vulnerability analysis and to protecting critical infrastructure: these links hold together otherwise separate parts of a network, and finding them in advance can pre-empt cascading collapse. Many traditional indices score edge criticality from a single, static view of the intact topology. But every edge removal reshapes the graph, so a one-shot ranking drifts out of step with the network it is dismantling. Deep Link Entropy and Improved Link Entropy showed that recomputing a metric after *each* removal works well, and we apply the same step-by-step re-estimation to three local edge-criticality metrics of increasing sophistication: Link Degree Centrality (LDC), Link-Local Betweenness Centrality (LLBCe), and Link-Local Betweenness Mapping Entropy (LLBMEe1). Using the relative-giant-component (RGC) curve and the area under it (AUC; lower is better) as the decomposition-efficiency criterion, we benchmark all three metrics, in both static and iterative form, on four real networks that span a wide topological range: the Jazz musicians collaboration graph, the Dolphins social network, a dense Baseball association graph, and a sparse urban Transport network. Iterative re-estimation improves LLBCe and LLBMEe1 on *every* network, by mean AUC reductions of 23.3% and 32.7% respectively and by up to 43.7%. For LDC the benefit is conditional: iteration helps clearly on the sparse Transport and Dolphins graphs but slightly degrades on the dense Jazz and Baseball graphs (mean 4.0% loss), a negative result we report rather than hide. Across all three metrics the gain from iteration grows as networks become sparser and more bridge-dominated.

Keywords: Complex Networks, Edge Significance, Edge Criticality, Link Degree Centrality, Link-Local Betweenness, Mapping Entropy, Iterative Re-estimation, Network Robustness.

1 Introduction

Complex networks underpin much of modern infrastructure. Examples include communication networks [6], power-supply networks [8], transport systems [9], and biological systems such as protein-DNA interaction graphs [1]. As more sectors come to depend on such networks, understanding how they withstand failure becomes necessary to keep them running.

Network science draws on theories and methods from several disciplines. A complex network can be examined through its topological structure or through the criticality of its individual links, and both views help make networks harder to fragment and pre-empt cascading failure. Researchers work in several directions: some predict new connections to optimise a network [1]; some rank influential *nodes* to assess resilience [2]; and some study critical *edges*, measuring the connectivity left after a link is removed. Robustness research has mostly concentrated on node removal, and

edge removal is comparatively under-explored. This study addresses that gap by removing edges rather than nodes.

Methods for evaluating edge criticality fall into two families. *Straightforward* algorithms analyse local topological features and assign a criticality value to each edge independently. They include Edge Betweenness [3], the Degree Product [7], Bridgeness [5], Diffusion Importance, Topological Overlap [6], k -path centrality, and Link Entropy. Their strengths differ by network type: the Degree Product does best on assortative networks, while Bridgeness does better on dense graphs. *Integral-optimisation* algorithms instead take a macro-level view; the Evolutionary Approach [12], for example, uses genetic principles to rank edges and can beat straightforward methods on ranking quality, but at a much higher computational cost.

Two recent straightforward algorithms, Deep Link Entropy [10] and Improved Link Entropy [11], both based on Link Entropy, improved efficiency through *iterative re-estimation*: recomputing the topological feature after each edge removal so the ranking tracks the changing network. We apply the same step-by-step re-evaluation to three local edge-criticality metrics of increasing sophistication and compare their static and iterative rankings:

- **Link Degree Centrality (LDC)**, a purely local degree-product statistic;
- **Link-Local Betweenness Centrality (LLBCe)**, edge betweenness restricted to the first-order central domain of the edge; and
- **Link-Local Betweenness Mapping Entropy (LLBMEe1)**, a metric that couples an edge’s local betweenness with the mapping entropy of its neighbourhood.

We numerically decompose four well-known benchmark networks chosen to span a wide topological range: the Jazz musicians network [14], the Dolphins social network [15], a dense Baseball association graph, and a sparse, near-tree urban Transport network. For each we compare the static and iterative variants under the relative-giant-component criterion. The rest of the paper introduces the three metrics and the iterative principle (section 2), describes the data, decomposition procedure, and per-network results (section 3), discusses cross-cutting patterns and limitations (section 4), and concludes (section 5).

2 Methods

This section introduces the three edge-criticality metrics (LDC, LLBCe, and LLBMEe1), then the iterative re-estimation principle applied to each, and finally the decomposition-efficiency criterion used throughout.

2.1 Link Degree Centrality (LDC)

Link Degree Centrality (LDC), also known as the degree product, scores an edge by the product of the degrees of its two endpoints. It originates from work describing cascading-failure-induced disasters in weighted networks [7]. For an edge $e = (u, v)$ in an undirected graph $G = (V, E)$,

$$\text{LDC}(e) = k_u \cdot k_v, \quad (1)$$

where k_u and k_v are the degrees of nodes u and v , that is, the number of edges incident on each endpoint. The idea is that an edge joining two high-degree hubs carries a large share of the network’s connectivity, so removing it should fragment the giant component faster. LDC is a purely local, $O(|E|)$ statistic and performs well on assortative networks, where high-degree nodes tend to attach to one another [7]. Edges are removed in *descending* LDC order.

2.2 Link-Local Betweenness Centrality (LLBCe)

Classical edge betweenness scores an edge by counting the shortest paths between all node pairs that pass through it [3]; an edge on many geodesics connects different parts of the network and is therefore critical. Evaluating betweenness over *all* node pairs is expensive. Link-Local Betweenness Centrality (LLBCe) restricts the computation to the *first-order central domain* of the edge, which keeps it sensitive to bridging structure while avoiding the global all-pairs cost [13]. The construction has three steps.

Step 1: central domain. For an edge $e = (u, v)$ define the first-order central domain

$$\Gamma(e) = \{u, v\} \cup N(u) \cup N(v), \quad (2)$$

the two endpoints together with all of their immediate neighbours. This local set delimits the node pairs over which betweenness will be accumulated.

Step 2: restricted path counting. For each ordered pair $s, t \in \Gamma(e)$ let σ_{st} be the number of shortest paths between s and t in the *full* graph and $\sigma_{st}(e)$ the number of those paths that pass through e . The ratio $\sigma_{st}(e)/\sigma_{st}$ is the share of s -to- t communication that relies on e ; geodesics are measured in the full graph, with no subgraph extraction.

Step 3: normalised aggregation. LLBCe is the average of these shares over the central domain,

$$\text{LLBCe}(e) = \frac{1}{|\Gamma(e)|(|\Gamma(e)| - 1)} \sum_{\substack{s, t \in \Gamma(e) \\ s \neq t}} \frac{\sigma_{st}(e)}{\sigma_{st}}. \quad (3)$$

Normalising by $|\Gamma(e)|(|\Gamma(e)| - 1)$, the number of ordered pairs in the domain, makes scores comparable across edges whose neighbourhoods differ in size. When the removal of e disconnects s and t the corresponding term is taken to be zero. Edges are removed in *descending* LLBCe order.

2.3 Link-Local Betweenness Mapping Entropy (LLBMEe1)

Link-Local Betweenness Mapping Entropy (LLBMEe1) combines the local betweenness of an edge with the *mapping entropy* of its neighbourhood, so it rewards edges that are both locally central and sitting in a heterogeneous local environment [13]. Let $\text{LLBCe}_{\text{raw}}(e)$ denote the un-normalised Link-Local Betweenness of e , the inner sum of eq. (3) before the $|\Gamma(e)|(|\Gamma(e)| - 1)$ division, and let $N(e_1)$ be the set of edges sharing an endpoint with e_1 . Then

$$\text{LLBMEe1}(e_1) = -\text{LLBCe}_{\text{raw}}(e_1) \sum_{e_2 \in N(e_1)} \log(\text{LLBCe}_{\text{raw}}(e_2)), \quad (4)$$

with the score set to 0 when $\text{LLBCe}_{\text{raw}}(e_1) \leq 0$ and the logarithm floored at $\log(10^{-10})$ for numerical stability. The metric combines two factors: the edge's own local betweenness, and the entropy-like sum of the log-betweenness of its incident edges, which is large when the neighbourhood is structurally diverse. The neighbour betweenness values are usually below unity, so their logarithms are negative and eq. (4) gives *negative* scores. The convention is therefore *inverted criticality*: the *most negative* edge is the most critical, and edges are removed in *ascending* order (most negative first). The sign is not guaranteed negative everywhere. If every neighbour's raw subset betweenness exceeds unity the sum of logs is positive, and a few edges can flip sign. We monitor this, but it does not change the ranking order materially on the networks studied here.

2.4 Iterative Re-estimation

In the conventional, *static* procedure a metric is evaluated once on the intact graph, edges are ranked, and they are removed in that fixed order. However, as the topology changes with each removal the criticality of the surviving edges may change as well, so the initial ranking becomes progressively stale. The *iterative* variant addresses this by re-evaluating the metric for every surviving edge after each removal and deleting the current most-critical edge, repeating until the graph is fully dismantled. The procedure is:

1. compute the metric for all edges of the current graph;
2. rank the edges and remove the single most-critical one;
3. record the size of the largest connected component and the fraction of edges removed; and
4. repeat from step 1 on the reduced graph until no edges remain.

This keeps the ranking aligned with the changing topology, the same principle that sharpened Deep Link Entropy [10] and Improved Link Entropy [11], at the cost of repeated metric evaluations. For LDC the recomputed quantity is the degree product; for LLBCe and LLBMEe1 it is the local betweenness (and its mapping entropy) over each edge’s first-order central domain, recomputed on the current graph.

2.5 Algorithm-Performance Measurement

A network is decomposed by removing its edges in criticality order, and the decomposition efficiency is read from the size of the largest connected component as edges are removed. Let $\rho \in [0, 1]$ be the fraction of edges removed and $R_{gc}(\rho)$ the size of the largest connected component relative to the original node count N . Initially $\rho = 0$ and $R_{gc} = 1$ (the graph is connected); when the network is fully decomposed $\rho = 1$ and $R_{gc} \rightarrow 0$. The more critical the edges removed early, the more abruptly R_{gc} falls as ρ grows, and the more efficient the ranking.

To summarise efficiency in a single number we use the area under the $R_{gc}(\rho)$ curve,

$$\text{AUC} = \int_0^1 R_{gc}(\rho) d\rho, \quad (5)$$

evaluated by the trapezoidal rule over the discrete removal sequence $\{R_{gc}(i/|E|)\}_{i=0}^{|E|}$. This criterion emphasises the global features of the decomposition rather than micro-level structure: a *lower* AUC indicates that critical edges were removed earlier and the network fragmented faster, i.e. a more effective ranking. Throughout, R_{gc} is normalised by the node count and ρ by the original edge count, so AUC values are directly comparable across networks of different sizes.

3 Data and Numerical Modelling

To compare the static and iterative variants of LDC, LLBCe and LLBMEe1 we apply the $R_{gc}(\rho)$ criterion to four real-world benchmark networks chosen to span a wide topological range, from dense, highly clustered social graphs to a sparse, near-tree spatial graph. All analysis is performed on the largest connected component with self-loops removed. Table 1 reports their structural statistics.

The four networks differ sharply in density and clustering. **Jazz** is a dense, highly clustered musician-collaboration graph; **Dolphins** is a small, sparse social network with a clear two-community structure; **Baseball** is an almost-complete association graph ($C = 0.977$, density 0.957);

Table 1: Structural statistics of the four benchmark networks (largest connected component): nodes N , edges E , average degree $\langle k \rangle$, maximum degree $k_{\max}$, degree heterogeneity $H_k = \langle k^2 \rangle / \langle k \rangle^2$, clustering coefficient C and edge density.

Network	N	E	$\langle k \rangle$	$k_{\max}$	H_k	C	density
Jazz	198	2742	27.70	100	1.395	0.618	0.1406
Dolphins	62	159	5.13	12	1.327	0.259	0.0841
Baseball	45	947	42.09	44	1.022	0.977	0.9566
Transport	369	430	2.33	7	1.190	0.029	0.0063

and **Transport** is a large, very sparse, nearly tree-like spatial graph ($C = 0.029$), where connectivity hinges on a few bridge edges. Figure 1 shows their circular layouts. In the following subsections we decompose each network with the static and iterative variants of the three metrics and describe the resulting $R_{gc}(\rho)$ curves; the consolidated AUC values appear in table 2 in section 4.

3.1 Jazz Musicians Network

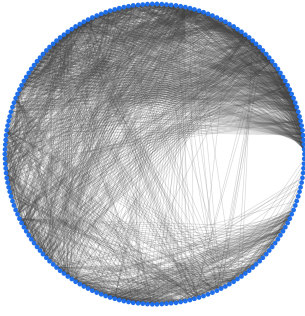
The Jazz network [14] captures collaborations between jazz musicians; its largest connected component has 198 nodes and 2742 edges, making it a dense, highly clustered medium-scale graph ($\langle k \rangle = 27.7$, $C = 0.618$). Figure 2 shows the static (solid) and iterative (dashed) $R_{gc}(\rho)$ curves for the three metrics. For both betweenness-based metrics the iterative curve falls earlier and steeper than its static counterpart: in such a dense graph each removal strongly rewires the local geodesic structure that LLBCe and LLBMEe1 depend on, so recomputation pays off, and LLBMEe1 improves more than any other network-metric combination in this study. For LDC the two curves nearly coincide and the iterative variant is slightly worse. On a graph this dense the degree-product ordering is already near-optimal, so recomputation just reshuffles edges of near-equal structural value.

3.2 Dolphins Social Network

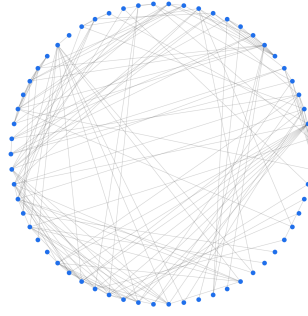
The Dolphins network [15] records 159 associations among 62 bottlenose dolphins of Doubtful Sound, New Zealand, and is a small, sparse social graph with a pronounced two-community structure ($\langle k \rangle = 5.1$, $C = 0.259$). Figure 3 shows its decomposition curves. Here iteration helps *all three* metrics: both betweenness-based metrics again improve substantially, and even LDC benefits modestly, because in a sparse community graph the degree product does become stale as the few inter-community links are exposed by successive removals. The iterative curves drop earlier, confirming that the adaptive ranking captures the bridging edges between the two communities more reliably than the static one.

3.3 Baseball Association Network

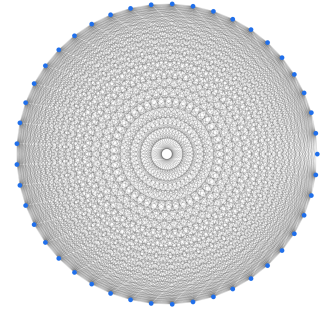
The Baseball network is an exceptionally dense, near-complete association graph of 45 nodes and 947 edges derived from a steroid-distribution case ($C = 0.977$, density 0.957). Because almost every pair of nodes is connected, the three static metrics produce nearly identical AUC values (all 0.936): when the graph is this close to complete, local betweenness, degree product and mapping entropy all rank edges almost interchangeably. Figure 4 shows the decomposition. Iteration transforms the two betweenness-based metrics, with LLBCe and LLBMEe1 each falling from 0.936 to roughly



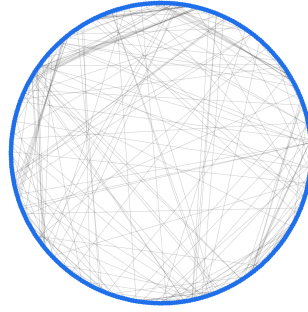
(a) Jazz



(b) Dolphins



(c) Baseball



(d) Transport

Figure 1: Circular layouts of the four benchmark networks. Density and clustering fall sharply from the near-complete Baseball graph and the dense Jazz graph to the sparse Dolphins social network and the near-tree Transport network.

0.67, because recomputation breaks the static ties and exposes the thin structural backbone that survives once the redundant edges are stripped away. LDC again resists iteration on this dense graph and worsens slightly, as it did on Jazz.

3.4 Urban Transport Network

The Transport network is a large, very sparse, almost tree-like spatial graph of 369 nodes and 430 edges ($\langle k \rangle = 2.3$, $C = 0.029$), where connectivity is governed by a small number of bridge edges. Figure 5 shows its decomposition. The absolute AUC values are the lowest of any network (the graph fragments abruptly once its bridges are cut), and iteration helps all three metrics, most of all LLBME1 and LDC: tracking the changing topology lets the iterative ranking target each cut-edge as it becomes critical, whereas the static ranking commits to an ordering computed before any of the tree-like structure has been exposed.

4 Discussion

Table 2 consolidates the static and iterative dismantling AUC for all three metrics on all four networks, together with the relative change Δ (positive = the iterative variant is more efficient). Three patterns emerge.

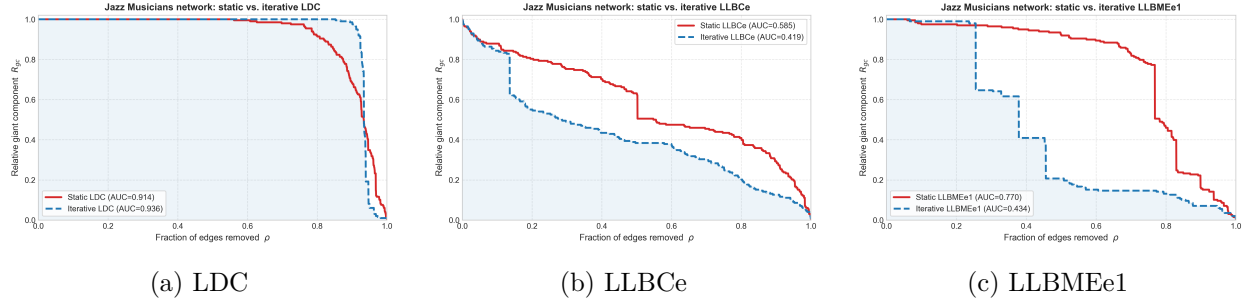


Figure 2: Static (solid) vs. iterative (dashed) dismantling curves for the **Jazz** network. Iteration sharpens both betweenness-based metrics strongly but does not help the degree product.

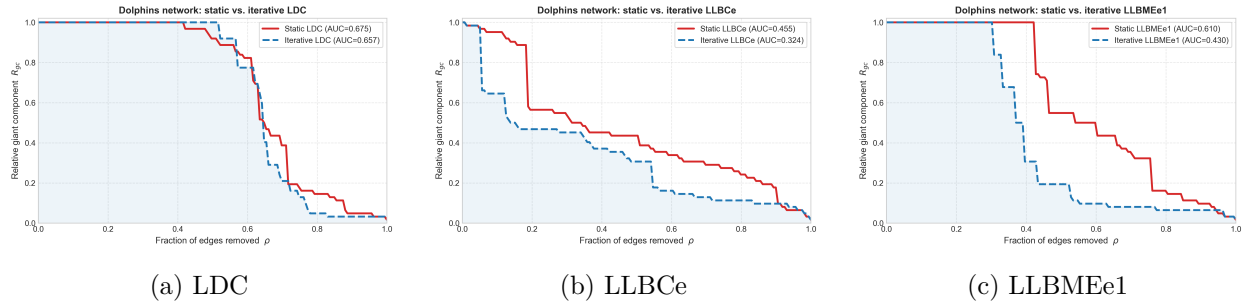


Figure 3: Static (solid) vs. iterative (dashed) dismantling curves for the **Dolphins** network. All three metrics improve under iteration on this sparse, two-community graph.

LLBMEe1 benefits most from iteration. The mapping-entropy metric improves on every network, by an average of 32.7% and by as much as 43.7% on Jazz. Because LLBMEe1 couples an edge’s local betweenness with the entropy of its neighbours’ betweenness, both factors shift sharply as the graph is dismantled, so a one-shot ranking quickly becomes stale and recomputation pays off most strongly.

LLBCe benefits consistently. Iterative LLBCe lowers the AUC on all four networks (mean 23.3%). The largest gains are on the dense social graphs (Jazz, Dolphins, and especially the near-complete Baseball graph, where iteration breaks the static ties), with a smaller but still positive gain on the sparse Transport network.

LDC benefits only on sparse graphs. The degree-product metric is the exception: iteration helps clearly on Transport (+18.1%) and modestly on Dolphins (+2.7%), but slightly *degrades* on the dense Jazz (−2.4%) and Baseball (−2.6%) graphs. On graphs that dense the degree-product ordering is already near-optimal and recomputation mostly reshuffles edges of near-equal structural value. We report this as an honest negative result rather than suppress it: iterative re-estimation is not universally beneficial, and its value depends on the interaction between the metric and the topology.

Topology governs the benefit of iteration. Across all three metrics the gain from iteration tends to grow as networks become sparser and more bridge-dominated (table 1). High clustering and density (Jazz $C = 0.618$, Baseball $C = 0.977$) provide local redundancy that blunts the effect of any single removal, so the static ranking remains serviceable; the near-tree Transport network

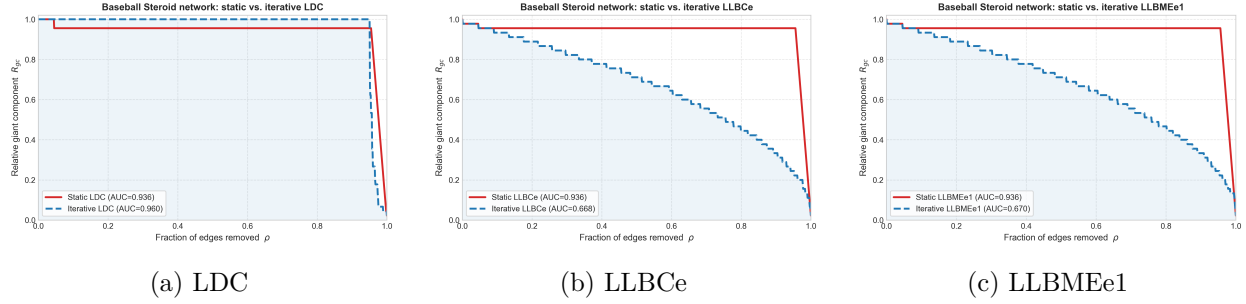


Figure 4: Static (solid) vs. iterative (dashed) dismantling curves for the **Baseball** network. On this near-complete graph the static metrics are nearly tied; iteration sharply separates the betweenness-based metrics.

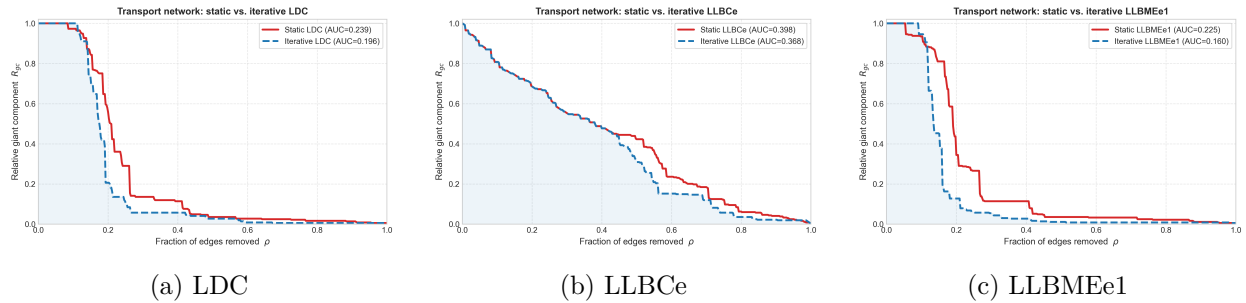


Figure 5: Static (solid) vs. iterative (dashed) dismantling curves for the **Transport** network. Iteration helps every metric on this near-tree graph, where connectivity hinges on a few bridge edges.

($C = 0.029$) instead fragments abruptly and rewards a ranking that continually tracks the changing topology. The one consistent exception to the “denser helps less” rule is the betweenness pair on Baseball, where the extreme density produces near-identical static scores that iteration is uniquely able to disambiguate.

Limitations. The principal limitation of iterative re-estimation is computational cost. Whereas a static ranking evaluates the metric once, the iterative variant recomputes it after every removal, multiplying the work by the number of edges; for the betweenness-based metrics, each recomputation itself involves repeated local shortest-path counting. Consequently the iterative variants scale far less favourably with network size than their static counterparts, and for time-sensitive decomposition tasks they may not be the optimal choice. A second limitation is the inverted-criticality sign convention of LLBMEe1 (section 2.3), whose log-sum can in principle change sign on edges whose neighbours all carry raw subset-betweenness above unity; we monitor this but did not observe it to alter ranking order materially on the four networks studied.

5 Conclusion

Edge criticality is a core problem in network science, and many methods address it across networks of different sizes and types. We took the iterative re-estimation principle, already shown to work for Deep Link Entropy and Improved Link Entropy, and applied it to three local edge-criticality metrics: Link Degree Centrality, Link-Local Betweenness Centrality, and Link-Local Betweenness

Table 2: Dismantling AUC (lower is better) for static and iterative LDC, LLBCe and LLBMEe1 across the four networks. Δ is the relative AUC reduction of the iterative variant; bold marks the two cases where iteration is slightly counter-productive.

Network	Metric	Static AUC	Iterative AUC	Δ (%)
Jazz	LDC	0.9139	0.9356	-2.37
	LLBCe	0.5850	0.4190	+28.38
	LLBMEe1	0.7701	0.4339	+43.66
Dolphins	LDC	0.6750	0.6571	+2.66
	LLBCe	0.4548	0.3236	+28.84
	LLBMEe1	0.6097	0.4298	+29.52
Baseball	LDC	0.9359	0.9600	-2.58
	LLBCe	0.9359	0.6684	+28.58
	LLBMEe1	0.9359	0.6704	+28.37
Transport	LDC	0.2390	0.1957	+18.11
	LLBCe	0.3981	0.3682	+7.51
	LLBMEe1	0.2251	0.1595	+29.10

Mapping Entropy. Using the relative-giant-component curve and its AUC as the decomposition-efficiency criterion, we benchmarked the static and iterative variant of each metric on four real networks, from dense social graphs to sparse, near-tree infrastructure.

The experiments showed that iterative re-estimation helps the two betweenness-based metrics, LLBCe and LLBMEe1, on every network tested, by mean AUC reductions of 23.3% and 32.7% respectively and by up to 43.7%. It also helps the degree-product metric LDC, but only on sparse, bridge-dominated graphs. The size of the benefit tracks network sparsity, growing as clustering and density fall. We do not aim to undermine the static methods, but to show that the iterative principle can add substantial benefit for certain local metrics in certain network contexts, while reporting honestly the two cases (LDC on the dense Jazz and Baseball graphs) where it does not.

Future work includes combining iterative re-estimation with metric-specific tie-breaking to recover the benefit on dense graphs and reducing the cost of repeated local-betweenness recomputation. Extending the comparison to larger infrastructure networks is a natural next step. These results point to when adaptive, topology-tracking rankings outperform their static counterparts, and they suggest further work on iterative refinements to classical edge-criticality metrics.

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Disclosure of Interests.

The authors have no competing interests to declare that are relevant to the content of this article.

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